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Gamification of Culture: A Strategy for Cultural Preservation and Local Sustainable Development

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Abstract: In the dual context of the cultural economy and the creative economy, culture is considered a unique source of competitiveness, making culture preservation a vital issue for peripheral areas. In this paper, it is argued that gamification is a valuable strategy for developing cross-cultural designs. The unique advantages of games in cultural preservation and promotion for peripheral areas with limited institutional resources are further articulated. To illustrate this, a set of design principles was proposed and the distinctive boat-based (Tanka) culture in southern China was taken as an example. A case study was conducted based on document analysis of the history of the Tanka culture and field research on the Tanka settlements in the Humen New Bay area, showing the development process and thus demonstrating the possibility of promoting the preservation and sustainable development of culture in peripheral areas through gamification. After testing the proposed game, some adjustments were suggested. However, participants generally agreed that they learned some Tanka customs through the gameplay, indicating the effectiveness of the game in Tanka culture preservation. This research thus contributes to both theory and practice by providing theoretical background and early practice.

Keywords: cultural and creative industries; CCIs; culture preservation; sustainable development; peripheral area; game; gamification; Tanka; southern China

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1. Introduction

Sustainability has always been a focal point for scientific research. The purpose of pursuing sustainability is to enable future generations to achieve our living standards or to live even better. Working toward that goal, it is crucial not to compromise the ability of future generations to meet their own needs [1]. Broadly speaking, in addition to physiological and psychological needs, the need for social interactions and/or cultural identification should also be considered. For people in the peripheral areas distinguished from the core by their contrastingly sparse population and lagging economic development [2], these needs are especially urgent through the lens of sustainability.

From the aspect of general economic development in modern society, the evolvement of various industries is closely related to the fulfillment of human needs, which are changing constantly and remain competitive. For example, manufacturing and service industries that both seek transformation within the knowledge and creative economies, as well as the cultural industries that emerge in the context of the cultural economy. In recent years, cultural and creative industries (CCIs) are increasingly recognized as of growing importance for sustainable local development [3,4], suggesting challenges induced and opportunities brought for the peripheral areas by the interactions between paradigms and domains of industry in terms of pursuing sustainable development.

On a macro level, from the aspect of the creative economy, culture is regarded as a unique source of competitiveness in the context of glocalization, making the preservation

of local culture a fundamental issue [5]. Taking this a step further, it is under the dual context of the cultural economy and the creative economy that the intertwined economic system and local culture in peripheral areas have emerged as a research topic, where there may be an opportunity to combine the preservation of cultural resources with local characteristics and the creative endeavors dedicated to local sustainable development.

On a micro level, along the line of sustainable development via CCIs, sustainability and culture have been widely discussed and can be categorized into three types of relations, namely culture in sustainability, culture for sustainability, and culture as sustainability [6]. Further distinguish can be made for the production and the consumption of cultural offerings. On the production end are people of a particular culture, for whom the cultural element embedded in the offering is their way of life and is anticipated to be sustainable until it is not. Hence, it is safe to say that, for people of a particular culture, the culture is culture for sustainability. Nonetheless, on the consumption end are people of another culture. For most of them, the cultural element in the offering is hardly a necessity; it is more of a complement and is considered an achievement in development, hence closer to culture in sustainability. For the peripheral areas where the sustainability of local cultures is threatened, it is important to recognize the different relations among culture and sustainability, so as to balance between "culture as a resource and condition" and "culture as a capital", and further seeking a strategy that takes cultural preservation and local sustainable development into account.

Taking the coastal area in southern China as an example, the population is concentrated in major cities, such as Guangzhou and Shenzhen, which impacts the economic structure and local culture of the smaller cities around them. Boat-based fishermen called the Tanka reside in these peripheral areas. The Tanka are not only distinctive in their way of life, but also have historical importance which has received academic attention, as they played vital roles in the interaction between China and the rest of the world in the 15th and 16th centuries [7]. However, as industrial landscapes change and ocean pollution increases, the marine fishing industry is threatened, and the Tanka people are urged to live and make a living on the land, resulting in the decline, if not demise, of the distinctive Tanka culture. In response to the potential demise of the Tanka culture, local governments launched relevant policies that support arts and crafts, organized cultural tourism, and held cultural experiences and activities but to a very limited effect, making visible the predicament of advancing cultural creative industries in peripheral areas. Approaching this issue from a sustainable development perspective could be of value.

Taking culture as the core, this paper discusses the major creative endeavors in contemporary cultural industries and points out that if cultural and creative industries are to be developed in peripheral areas with generally limited institutional resources, cultural innovation has higher sustainability, while the unique advantages of a gamified cultural product in cultural preservation and promotion are articulated and further emphasized. With a set of design principles proposed, a case study of a Tanka culture game is described, showing the development process and thus demonstrating the possibility of promoting the preservation and sustainable development of culture in peripheral areas through gamification.

2. Entangling Streams of Culture, Creativity, Economy, and Industries

The concept of "cultural economy" focuses on analyzing the production, distribution, and reception of symbolic content [8]. This term can be vague as it is polysemous while overlapping with creative industries and cultural creative industries, but it undoubtedly represents a dynamic trend of thinking, implying multi-faceted innovation opportunities such that it may contribute to economic development and structural transformation, as well as regional development and environmental sustainability [9–12]. With cultural preservation and local sustainable development as the core objectives, this paper does not distinguish concepts of cultural economy, creative economy, cultural industry, and cre-

ative industry but uses the umbrella term "cultural and creative industries (CCIs)" for related research.

Research on cultural creative industries can be classified into three stages [13,14]. Each stage leads to a corresponding trend of research. In the first stage, attention is paid to the economics and production of culture; in the context of the cultural economy, cultural content is regarded as the essential fuel for economic development. The positive impact of cultural investment on economic growth has also been acknowledged [15,16]. With cultural industries being regarded as an important component of the economy, the culture-centered perspective has a profound impact on policy formulation and industrial development. Here, the concentric circle model proposed by Throsby (2008) [17] is one of the widely known theories, with literature, music, and the performing arts taken as the core creative arts, which extend to cover related industries, such as design and cultural sightseeing, to build a concise system concerning the cultural industry [17,18]. The core concept of culture has been explored in depth by the academic community. For example, Sacco, Ferilli [19] analyzed the evolution of the relationship between cultural production and different forms of economic and social value creation in terms of three alternative socio-technical regimes that emerged over time. They also described the development from Culture 1.0 to Culture 3.0 and emphasized shifting the focus from the economic outcomes of cultural activity to the behaviors that cause them [19].

In the second stage, the focus shifts to regional development and the relationship between CCIs and the economy as a whole and, in particular, cultural policy, the geographic economy, and employment [14], which can be traced back to the attention toward the creative classes in the urban area near Florida [20], including how the creative classes affect the region's economic growth [21–23], the migration and living preferences of the creative classes [24,25], and the attempts to figure out the essence of creative labor and how to manage it [26–28]. In this direction, the heterogeneity and internal fragmentation of CCI research continue to grow [29], which can be attributed to the differences in research background and setting of the individual research. Though these reasons can be put under the umbrella term of "location factor", which is undoubtedly crucial for the development of CCIs, the umbrella term also covers such aspects as the institutional environment, policy making, and industry operation, where differences may exist. Therefore, if sustainable development is considered a valuable aim, creative endeavors should be channeled in a targeted way [3,30].

The third stage focuses more on regional variations, peripheral places, and urban regeneration [13,14]. For example, regarding the creative industry's impact, Bontje and Musterd [31] made a critical appraisal of local experts' views on developments in seven European city regions. Jayne, Gibson [11] found related studies on the cultural economy of small cities, pointing out that small cities and their regions are also shaped and changed by different regulatory arrangements and sociocultural imaginaries. However, as unique factors, such as population, size, and proximity, cause pressure as well as bring opportunities to small cities, research about them can contribute to a more inclusive framework [11].

Taking a step back and viewing the stages of CCI research, it can be observed that a merging between culture-centered and industry-oriented approaches is in progress (as depicted in Figure 1). Amid the general premise of advancing prolificacy via technological progress and its latent concern toward human well-being, sustainability for peripheral areas has become an emerging theme, whereas cultural innovation, especially in tangible forms, has great potential and awaits exploitation.

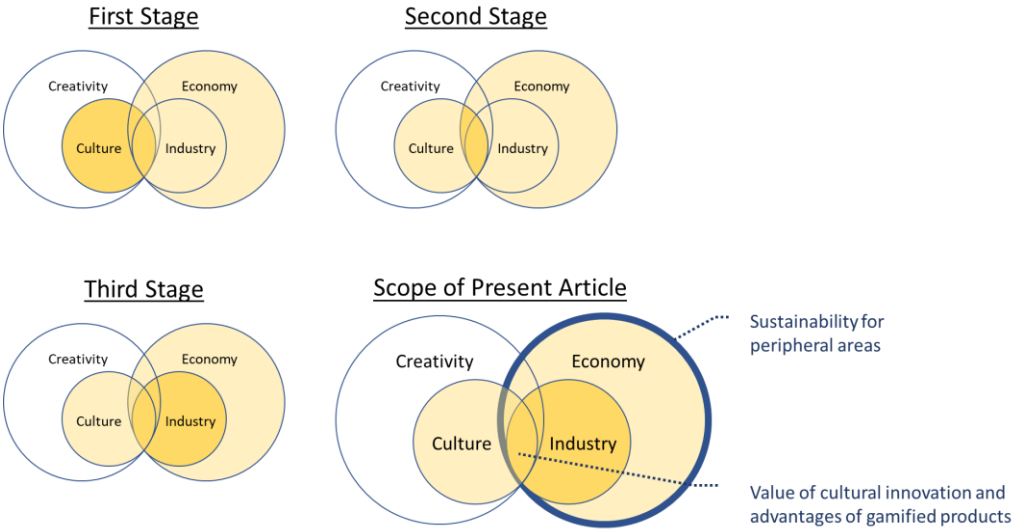


Figure 1. Stages of CCI research and the scope of the present article.

2.1. Sustainability for Peripheral Areas via Cultural Innovation

Whether from the viewpoint of CCIs or glocalization, culture is undoubtedly a core resource for innovation and local sustainable development. Additionally, activities linked to cultural heritage (cultural and educational but also conservation and restoration activities) addressed to different social groups are currently regarded as a chance of social inclusion of persons and social groups in danger of marginalization [32], which could further enhance social capital for peripheral areas. In this vein, when considering the sustainable development of peripheral areas, local culture with unique characteristics should be considered as the source of distinctive competitiveness. Hence, it is important to prioritize the inheritance of culture in peripheral areas [33–35] and consider the intertwined economic system and local culture in those areas to determine a suitable strategy for fostering cultural innovation.

In this vein, we can state the possible methods to achieve cultural innovation and further categorize the forms of innovative outputs by CCIs into "cultural tourism" oriented toward service industries and "cultural products" oriented toward manufacturing industries. Cultural tourism attracts visitors to places mainly through the cultural experience at those destinations or during events and makes people spend money at the destinations [36]. Successful or sustainable cultural tourism entails infrastructure, such as accommodation, parking spaces, meeting rooms, and other subsidiary facilities [37,38]. However, peripheral areas are often lacking in infrastructure and have insufficient coordination between resources and assets, which makes it hard for them to provide stable and well-rounded support for tourism services, and thus, this negatively affects the service quality [39–42]. For example, many local stakeholders in Kemi (Finland) expressed their concern regarding the existing hotels which many of them described as outdated and inadequate for tourism [43]. In another case of Almeida (Portugal), besides limited infrastructure and public facilities, this peripheral area also faced with human resources constraints, such as ageing population and lower education levels, and is considered an area in economic decline where there are limited alternative paths to development and sparse endogenous resources to invest [44]. In contrast, the development of cultural products is based mainly on the manufacturing industry, where cultural assets are converted into cultural products, such as physical objects, TV programs, films, or entertainment, and are sold to other places [45–47]. Under these circumstances, viable cultural products may be developed in accordance with the manufacturing resource conditions in the peripheral areas.

Another axis of discussion exists in terms of manufactured products. Concerns regarding cultural meanings or implications are essential for cultural innovation, while functionality is core and unavoidable for most contemporary product development. In many

cases, the inconsistency between the product's functionality and its cultural elements renders the design of cultural products difficult due to its complexity. As compared to utilitarian value latent to functionality, the risk of inconsistency is relatively low for products addressing aesthetic or hedonic value, such as games and toys, providing a potential remedy for the peripheral areas.

Toys and games, with their meanings in play and sometimes in earnest, can be seen as aspects of communication and the transmission of culture [48]. Such hedonic products have been adopted. For instance, Torres and Macedo (2000) developed a card game to create awareness of and explore attitudes toward environmental conservation and urban development [49]. More recently, Kuntjara (2022) found that art toys based on the city's symbols were a potential platform and asset for engaging the public for Surabaya in Indonesia [50].

In a societal sense, the emphasis on hedonic products echoes the idea of "tourist as child" as articulated by Dann and Cohen [51], who viewed participants of a touristic experience as reverting to a childlike stage of existence and having their spontaneous hedonism checked by the industry through its language of social control. In a business sense, from a broader aspect of the experience economy, hedonic value is regarded as a key aspect of the consumption experience [52,53]. Hence, as one major category of design outcomes that are producing strategies for development, games and gamification that mainly deliver hedonic value provide a unique approach that deserves in-depth discussion.

2.2. Gamification of Culture

As a kind of simulation in which consumers willingly participate, games have the potential for changing the attitude and stimulating the motivation of participants [54–56]. Gamification means introducing game mechanisms into a non-game context or redesigning activities in life by drawing inspiration from game design [57,58]. At an operational level, from the perspective of the service system, gamification can be described as a process of enhancing a service with affordances for gameplay experiences to support users' overall value creation [59]. Over recent years, gamification was widely applied in education and training, and its application in the cultural industry was seen in the protection and promotion of cultural heritage [60,61] and festivals and events [62,63], with the focus mainly on tourism and cultural experience and less attention being paid to applications in product development.

For learning, preserving, and promoting culture, the convergence of education and entertainment could bring unique advantages [64] and can be further discussed through the closer inspection of games. The constituent elements of a game include the goal, means, rules, and lusory attitude, in which rules play a prescriptive role [65,66]. One of the important features of game-playing is the players' appropriate epistemic access to goals and rules and commitment prior to the start of the game so that they can be guided by the game's goals and rules [67–70]. In other words, if we translate cultural behavior, such as a certain ceremony, into a game, when people decide to "play the cultural game", it means that they understand and accept the goal of the cultural behavior and are willing to conform to the inherently cultural rule. Therefore, the game becomes a useful tool that channels consumers to break free from the boundary of their own culture, making it easier for them to accept the unique existence of other cultures. This could lead to their further appreciation of the development or decline of such cultures, contributing to culture preservation in peripheral areas.

2.3. Opportunity for Tangible Gamified Cultural Products in the Emerging Orange Economy

From the above arguments, it was proposed that turning cultural content into the form of a tangible product with hedonic value, i.e., games, could be valuable for a culture's preservation, namely, the "gamification of culture." Thus, gamification is considered a strategy for pursuing local sustainable development in peripheral areas and could be of value to the peripheral or even broader areas generally referred to as the Global South.

In Latin America and the Caribbean, the concept of the "orange economy (OE)" has been increasingly perceived as an opportunity for growth. The orange economy (OE) is defined by the Inter-American Development Bank (IDB) as "the set of activities that in an interlocking way allow for ideas to be transformed into cultural goods and services" [71]. A close relationship and resemblance can be observed between the OE and CCIs, as they both acknowledge the central role of creativity for economic growth. However, the OE's developmental path and potential trajectory could be complementary and of value to CCI research on peripheral areas.

The OE has been increasingly recognized for its potential to create employment, increase cohesion, promote innovation, and contribute to the well-being of societies, as it assumes both commercial and cultural values. It has also been recognized for its relevance toward achieving the Sustainable Development Goals (SDGs) [72,73]. Results showed that the OE is a sector that is resilient against crisis [74]; it could also be a tool that offers the necessary instruments to enhance the value of the locals [75]. In general, the OE is considered an engine and a generator of sustained growth over time [74]. This helps to keep sustainability in focus for CCI research, especially in areas where resources are in short supply and the economic system is vulnerable.

The OE is distinctive in that it is still from a more bottom-up perspective in which governmental bodies try to harness the endogenous potential of local society. In comparison, regardless of the industries covered, CCIs were also a more bottom-up approach when first proposed by the UK's Department for Digital, Culture, Media and Sport (DCMS) in the 2000s. However, as mentioned at the beginning of Section 2, the research focus gradually shifted to the economy, which many governments constantly try to administer via top-down policymaking. This channels academics into an administrative mindset and top-down approaches such that when they are trying to create an impact on the real world, the structural forces that made the peripheral areas peripheral often emerge as barriers and hinder the implementation of research results. In this regard, the emergence of the OE bringing a refreshing bottom-up perspective reminds the researcher to look at those successful cases that can neither be achieved nor explained by top-down approaches.

The proposed "gamification of culture" approach is essentially a bottom–top approach and requires exploring the cultural content that could be converted into the form of a tangible game. Accordingly, it deviates from major streams of CCI research and coincides with the context of the OE, bridging the two realms of research and contributing to the discussion of preserving culture in peripheral areas.

3. Gamification Strategy for Cultural Preservation and Local Sustainable Development

3.1. Principles of the "Gamification of Culture"

In guiding developing culture innovation in this study, the structure of Mechanics, Dynamics, and Aesthetics (MDA) was adopted with the principles of the "gamification of culture" corresponding to its structural parts.

As an approach to understanding games, MDA provides a useful model for us to grasp how gamification works [76]. In MDA, the consumption of games is formalized by breaking the game into distinct components, namely mechanics, dynamics, and aesthetics, and describing the data representation and algorithms, the run-time behavior of the mechanics, and the desirable emotional responses evoked in the player when they interact with the game system [77].

Additionally, modifications and replacements of MDA were subsequently proposed by various scholars and are of potential value to gamification in CCIs as a whole, including the Design, Dynamics, Experience (DDE) framework [78], which is largely referenced in this research. As the three-pillar structure of MDA remains and it is beyond the scope of this research to discuss different or synthesis models inspired by MDA, the original terms for the corresponding MDA parts were adopted when referencing other MDA-like models for game design. It is also worth noting that, at least at this stage, the principles presented

in this paper serve only as part of the development process of a culture product and should not be seen as any generalized formative meaning.

The principles of the "gamification of culture" are as follows:

1. Aesthetic: Cultural experience is the core of cultural innovation. Therefore, when extracting cultural content, designers should have a clear goal of what to preserve.
2. Mechanism: A fundamental belief is that a game mechanism is a simplified extraction of the real world when being developed for culture preservation. Therefore, instead of developing a game mechanism from scratch, designers should search for a mechanism that channels user behaviors toward the selected cultural content.
3. Dynamic: Designers must be careful with the game dynamics when finetuning the game, and the gameplay must be smooth and fun.

3.2. Cultural Content as Design Material

When referring to potential cultural resources in CCIs or related fields, there exist concepts covering ideas of similar range but with different interests, which might lead to confusion, namely "cultural patrimony", "cultural heritage", and "cultural content". To clear distinguish between the concepts is beyond the scope of this research, however it is only ethical to discuss on the terms that coincide more precisely with the proposed gamification strategy.

The term "cultural patrimony" refers to that part of a national, tribal, or other society's culture, which is so fundamental to the society's identity and character that people deem it inalienable [79,80]. The lack of cultural patrimony can pose a risk to the preservation of cultural heritage, and it is important to develop innovative business models which guarantee sustainable growth in line with market forces [81]. Since the 19th century, a developing awareness of cultural patrimony has been accompanied by the growth of a body of literature on conservation, and by the establishment of agencies at the international level, dedicated to conservation [82]. On the other hand, "cultural heritage" is the legacy of physical artifacts and intangible attributes of society inherited from past generations and is regarded as old, important, and worthy of conservation [83,84]; it may include physical artifacts, such as works of art, literature, music, archaeological, and historical artifacts, as well as buildings, monuments, and historic places. Meanwhile, intangible attributes comprise social customs, traditions, and practices often grounded in aesthetic and spiritual beliefs and oral traditions [84]. Although the importance of immateriality is being acknowledged, making intangible heritage an emerging target of concern, there is no denying that tangible heritage still constitutes a great part of the concept, and the connotation of materiality remain strong for the term cultural heritage, at least for now. In contemporary usage, both "cultural heritage" and "cultural patrimony" are synonyms, metaphorically depicting the idea that the culture (material and immaterial) of a specific social group is its property, owned collectively and passed on from one generation to the next [85], and less considered as something that is to be conveyed to people outside of the culture zone.

In contrast, the term "cultural content" refers to the symbolic meaning, artistic dimension, and cultural values which originate from or express cultural identities [86] and is often regarded in relation to media and communication. In addition, the term is used to promote diversity in education. Such a concept involves the integration of examples, concepts, perspectives, and the lived practices of diverse cultural and ethnic groups into the mainstream curriculum and schooling processes [87]. As the UNESCO world report dedicated a whole chapter to the landscape of communication and cultural contents, it further pointed out that today's communication and cultural products are becoming powerful tools of non-formal education and cultural transmission with the potential to expand intercultural understanding [88]. From such a viewpoint, with the connotation of communication, cultural content would be a term more in line with gamification strategy.

Aimed at the distinctive boat-based (Tanka) culture in southern China, this study involved first sorting through the history of Tanka culture based on document analysis and then conducting field research on the Tanka settlements in the Humen New Bay area

(虎門), which illustrated Tanka culture preservation while collecting material for cultural game development.

The Tanka or the Soi-Seong-Ian (水上人), which means "people on the water", living along the southern coast of China, are the descendants of a pre-Han-Chinese autochthonous tribe [89]. The Tanka live off their boats, and their population is hard to discern. More recent studies on the population of the Tanka are mostly scattered at the municipal level and show the widespread habitats of the Tanka. Nonetheless, according to an earlier survey conducted during WWII, among the 32 million people in Kwangtung (廣東), the Tanka had a population of 3 million [90]. The Tanka, whose main economic activity is fishing, have throughout their history been compelled by the Chinese to live in isolation [91]. As a result, they have developed unique cultural and trade activities. In the Pearl River Delta region, they were the first group of people to have close contact with the Europeans [7].

In recent years, the Tanka have moved to the shore, either autonomously or on a policy basis, which, while improving their living conditions, has caused anxiety about their change in lifestyle and concern about the disappearance and preservation of related culture [92–94]. Based on existing literature, this study conducted field research on the Tanka settlements in the Humen New Bay area, Dongguan city, and Guangdong province, and summed up the main aspects of Tanka culture.

3.2.1. Food

The Tanka mainly live by the sea, and thus, their basic edible ingredients are obtained from the sea (see Figure 2). As they always drift on the sea, their food is scarce and monotonous. In cooking, the Tanka seldom fry or deep fry but put the food in boiling water, which retains the original delicious flavor of the food, such as fresh shrimp and tender fish. As the wind on the sea is very strong, the Tanka also dry the food they have hauled out of the sea. They spread the slices of seafood on a bamboo container, where the food is made delicious under the sea wind and sunshine.



Figure 2. Tanka fishing boats in Humen; the red flags in the right-hand image indicate that there is a fishing moratorium.

3.2.2. Clothing

The clothing culture accentuates the ethos and material culture of a nation. With changes in the crowd and the environment, the clothing culture is also improving. Given the living environment and characteristics of the Tanka, as they have been drifting on the sea, the Tanka wear clothes that have the color of the sea. Their clothes are mostly blue and made of cloth, with the cuff of the jacket being wide and the sleeves short; the Tanka mainly wear shorts. In addition, the Tanka wear the distinctive Tanka hat, which is mainly made with bamboo strips and bamboo slices. Overall, the clothing of the Tanka is relatively simple, which not only evokes a feel related to the sea, but also serves practical functions.

3.2.3. Habitation

For the Tanka, the sea is their land and the boat is their house. The Tanka fishing boats differ in style. Compared with a traditional boat, the Tanka fishing boat has a relatively short and mostly oval hull with unique decoration.

3.2.4. Marriage

In Tanka culture, Tanka songs and dumplings in tea soup constitute the core of the Tanka marriage custom. The "dragon boat"-style wedding involves a representative folk dance and folk art created through their long-term labor, and the wedding ceremony is very festive (see Figure 3). Everything from the props to the actions to the costumes has rich cultural connotations. On the wedding day, more than 20 middle-aged women from the fishing village, dressed in self-made clothes and hats, form the shape of a fishing boat on the shore, seeming to work to the beats of the dragon boat and head to the bride's house. With a black umbrella hung with red streamers in hand, the groom stands at the bow of the boat together with the bride as the boat gradually moves toward the groom's house. In Tanka culture, Tanka songs also constitute an important component. All parts of the marriage custom carry the traditional features of the Tanka people and show that the Tanka love singing. Their songs represent not only a part of folk music, but also a type of folk music with special tunes and rhythms.



Figure 3. Tanka marriage custom as exhibited in the Tanka museum in Humen (新湾疍家文化展示馆).

3.2.5. Music and Literature: Tanka Ballads

The iconic songs that the Tanka most like to sing are Tanka ballads (see Figure 4). In daily life, the Tanka always hum such songs, which mostly describe their harsh life, as well as the love between men and women. On the whole, the songs are steady in rhythm, with the stanza repeated in each section. The Tanka songs not only bring out the Tanka's delights and lifestyle, but also enable people to relax and bring them closer to each other. In the Tanka marriage custom, the ballads play an important role and can be said to be representative of the Tanka culture. With the decline in the Tanka culture, the orally transmitted Tanka ballads have become a target of preservation [95].



Figure 4. Tanka ballad about a wife wishing for the safety of her husband; the photographs were taken at a local residential center in Humen (reproduced with the permission of Jia-Ting Xiong (second to the right on the right-hand image)). Rough translation of the ballad on the left image is as follow: *Large boats line up in port. Leaving her basket expecting her man, she hired a raft and closing. Many men yet none of them. She asked the boat hand. She asked some young men. To the crew of her man, she asked. You go as one yet come without mine, why? 'He was on the boat,' the crew said. 'Hainan as we get, a girl he met,' bangs in front and bun on her head, neither tall nor short, pointed fingers with delicate toes. 'a lovely girl he met'.*

4. Culture Game Development: The Tanka Ballad Game

Taking cultural content as the core and a game as the medium, Tanka ballads sung as part of a wedding ceremony were transformed into a game in this study. The game involves players being at a Tanka wedding ceremony where they seek partners with a matching token and further collect lyrics and jointly complete a ballad, almost like they were singing a ballad together.

4.1. Game Structure and Victory Goal

A core aim of singing Tanka ballads is for pairing, with both sides singing the ballad to confirm the feelings of each other. Based on such a cultural phenomenon, the game has the victory goal that two players pair up and complete lyrics for a ballad. Therefore, the game adopts "set collection" as the core mechanism for the victory goal, and a "cooperative game" structure is adopted, with an even number of players being required. Additionally, the game supports a total of four to eight players for a better pace.

4.2. Mechanisms and Rules

The core mechanism for the game is "hand management", which means players try to gain the most value out of available cards by playing cards in certain sequences or groups under certain circumstances. Further, some randomness mechanisms are introduced into the game to increase variation in the gameplay, including card draw and limits, variable set-up, and events.

In summary, the game consists of two stages (see Figure 5). First, a pairing stage where players try to find a partner by matching tokens. The matching mechanism and rules are as follows:

1. At the beginning of the game, each player is dealt four token cards.

2. A starting player may draw a card from their neighbor. If the player picks a card with the same token as another card in their hand, the matching pair can be claimed, and the two players form a team. The rest of the cards in their hand go to the discarded pile. The drawing of cards continues in turn until all players have found a match. For example, if Player 1 draws a card from Player 2 and picks a token card that is the same as a card in Player 1's hand, Player 1 can claim the matching pair and form a team with Player 2; if there is no match, it is Player 2's turn to draw a card from their neighbor, and the game goes on like this.
3. When all players are paired up, the game goes on to the next stage.

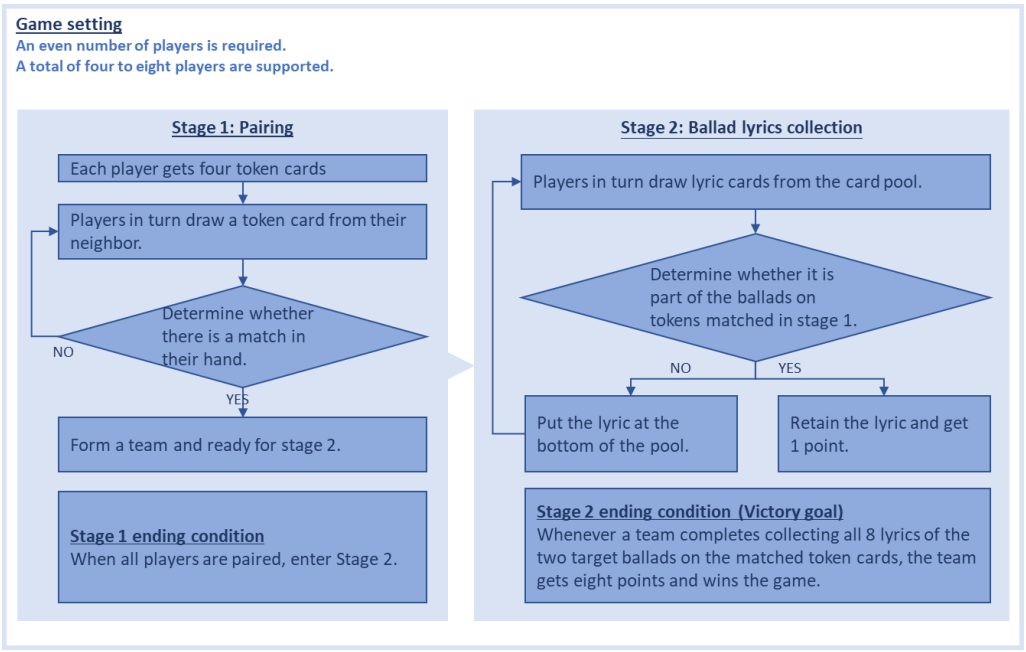


Figure 5. Game stages and flow.

The second stage involves collecting ballad lyrics. For each token card, there is a Tanka ballad to signify the goal for team victory when the tokens are matched. For each team, there are two token cards, and therefore, two ballads. The players draw in specific sequences and act according to the rules specified, as follows:

1. The first paired team is the first to draw lyric cards, as they were ahead in the matching process. The rest of the teams draw according to the sequence of success in pairing or determined jointly by all players.
2. During the card drawing, when a player picks one lyric that belongs to the target ballads, the player receives one point; if they do not pick a target lyric, nothing happens, and the lyric goes to the bottom of the pool. In maintaining balance in the game, each player can have no more than seven cards in their hand, and any extra ones are discarded.
3. The pool also contains event cards that introduce related marriage customs, providing players with events or actions that the player must carry out immediately or in the following rounds as prescribed on the cards.
4. When a team collects all eight lyrics of the two target ballads on the matched token cards, the team receives eight points and wins the game. The formats for token cards, lyric cards, and event cards can be seen in Figure 6.

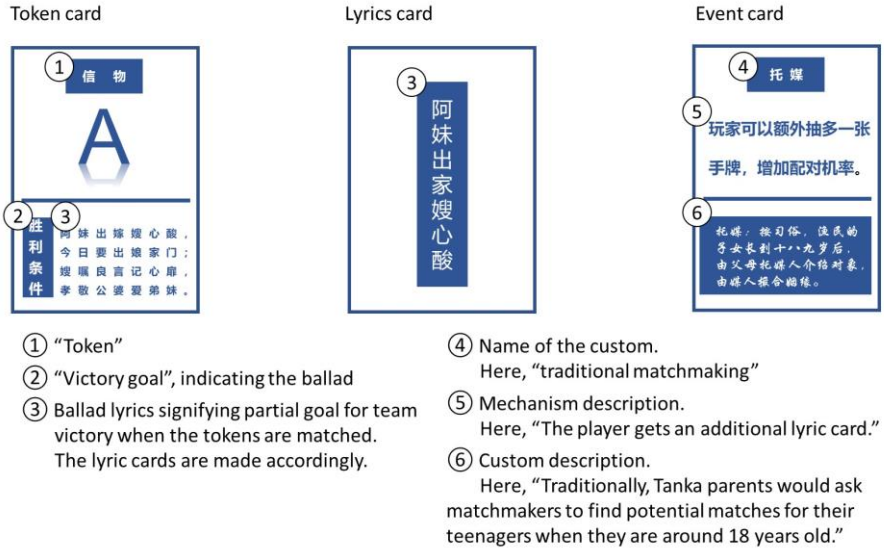


Figure 6. Formats for token cards, lyric cards, and event cards.

4.3. Demonstration of the Tangible Gamified Cultural Product

In testing and demonstrating the playability of the game, a group of participants who were not familiar with Tanka culture was invited to play the game in repeating rounds and a focus group interview was conducted. The participants were three males and three females aged between 20 and 30 years.

As a result, three major findings were obtained from the aspects of game design and culture preservation. As for the game design, on average, the game took a long time to play; however, an increased game pace was expected. A major reason lay in the matching process, as players had to wait for others to complete matching and often felt bored, suggesting the need for a matching mechanism that is either short and efficient or invigorating but not necessarily in line with the Tanka marriage custom. In terms of culture preservation, participants generally agreed that they learned some Tanka customs through the event cards and could remember some lyrics of the Tanka ballads, suggesting effective culture learning at least in some form. The participants further suggested that players should be asked to sing the ballads with their pronunciation and rhythm being taught or even incorporated into the game mechanisms.

5. Discussions

This research provided two layers to be discussed: one is on the design outcome, focusing on the value of the Tanka ballads game; another is on the project, focusing on articulating how gamification could be a feasible strategy for culture preservation and could benefit the sustainability of peripheral areas.

5.1. Value of Tangible Culture Product for Peripheral Areas

In the presented case of Tanka culture, as mentioned above, existing efforts have been made by governments and NGOs in preserving the culture, as signified by the establishment of the committee (or working group) for Tanka culture preservation (疍家文化保育委员会或工作組) supported by municipal governments in various regions. More specifically, in the Humen New Bay area, a docent training program for teenagers (青少年讲解员公益培训活动), a variety show (疍家文化特色文艺晚会), family activities (疍家文化亲子研学), and a culture festival (疍家文化节) have been implemented.

As can be observed, the current efforts focus more on promoting cultural tourism, but the result may not be all that satisfying. In a brief comparison of the growth rate of inbound visitors before the COVID-19 pandemic (as shown in Table 1), the performance of Dongguan city was behind that of the whole of Guangdong province, indicating that

cultural tourism in Dongguan might not be growing very fast, suggesting that there is room for additional effort in promoting local culture, which may include the Tanka culture in Humen.

Table 1. Total number of inbound visitors and growth rate before the COVID-19 pandemic.

Year	Guangdong Province [1]	Growth Rate (%)	Dongguan City [2]	Growth Rate (%)
2016	397,183,900	-	37,919,755	-
2017	443,475,600	11.65%	41,418,524	9.23%
2018	488,078,293	10.60%	44,338,730	7.05%
2019	531,410,169	8.41%	47,486,106	7.10%

[1] Data retrieved from the Bureau of Statistics of Guangdong province (http://stats.gd.gov.cn/) (accessed on 9 December 2022). [2] Data retrieved from the Bureau of Culture Affairs, Media, Tourism and Travel, and Sport of Dongguan city (http://wglt.dg.gov.cn/) (accessed on 9 December 2022). As data from the Humen New Bay area were not available, the larger area of Dongguan city was adopted for comparison. No data were available from before 2018.

To be more explicit, the benefit of tangible gamified cultural products is two-fold, as articulated with the presented case of the Tankas in Dongguan. The first is concern with the development of industry from a resource-based perspective, as the impact of resource constraints is less profound for the tangible product compared with a service.

Following the argument in Section 2.1, cultural tourism that preserves culture through visitor experiencing in the local area is confronted with the fact that ordinary service capacity is inevitably limited in the peripheral areas. This further lead to concerns on human resources and social equality regarding tourism as it could be more vulnerable to environment volatility, such as the outbreak of the COVID-19 pandemic [96]. While women, youths, and low-income, low-skilled groups were found to endure disproportionally greater economic hardship with higher unemployment rates and higher pay cuts than their male counterparts during the COVID-19 pandemic, it is important to note that significant differences in job vulnerability are observed across regions, sectors, and sociodemographic groups and this could be a result of local economic structure and the composition of the labor force [97]. This issue would be difficult to tackle without more thorough investigation across the area, which would be resource-consuming and challenging for peripheral areas.

For the presented Tanka case in Dongguan, Guangdong, as seen in Table 2, the number of star-rated hotels is decreasing, suggesting a shrinking service capacity that could impact visitor motivation, as seen in Table 3.

Table 2. Number of hotels in Dongguan City [1].

Year	1-Star or Above Hotels	4-Star Hotels	5-Star Hotels	Total Beds
2019	29	11	13	13,360
2020	26	10	12	12,078
2021	22	7	11	9096

[1] Data retrieved from the Bureau of Culture Affairs, Media, Tourism and Travel, and Sport of Dongguan city (http://wglt.dg.gov.cn/) (accessed on 9 December 2022).

Echoing the discussion on sustainability for peripheral areas Section 2.1, as the goal is to preserve and promote local culture, tangible products present a feasible strategy for peripheral areas. Many decisions would be more flexible, from the decision on product category, to its materials and production method.

Table 3. Total number of inbound visitors, growth rate, and tourism revenue of Dongguan City [1] in 2019–2021.

Year	Domestic (Growth Rate)	International (Growth Rate)	Total Revenue
2019	43,436,255 (7.73%)	4,049,851 (0.76%)	5,741,640 (8.46%)
2020	38,517,004 (−24.86%)	248,269 (−80.29%)	3,586,295 (−30.17%) [2]
2021	45,306,822 (17.63%)	228,784 (−7.85%)	3,795,765 (5.84%) [3]

[1] Data retrieved from the Bureau of Culture Affairs, Media, Tourism and Travel, and Sport of Dongguan city (http://wglt.dg.gov.cn/) (accessed on 9 December 2022). [2] In the 2020 report, total revenue for tourism in 2019 is 5,135,435 and not 5,741,640 as in the 2019 report. If calculated based on data originally reported in 2019, the growth rete is −37.54% in 2020. [3] A bounce back can be observed mostly on number of domestic visitors, yet shown limited contribution to total revenue for tourism as comparing to 2019.

The second benefit is concern with the potential market as from industrial organization perspective. The unique culture of the peripheral area can undoubtedly provide a source of differentiation and potentially give it a competitive edge. Moreover, the potential revenue for a gamified product could be vast. Based on other supplementary data provided by staff in the Tanka museum in Humen, between 2018 and 2022, around 80 groups visited with an accumulated visitor number of around 30,000. Considering that games are major types of crowdfunding projects in China [98], that number is relatively small when taking into account China's large domestic market and sophisticated value chain for tangible products. One the whole, the advantage of a tangible culture game for peripheral areas is almost self-evident for countries with a vibrant domestic tourism market.

In summary, the possibilities for the incorporation and study of culture and gamification are vast and can be explored via cultural values that are planned and enhanced by the design of an interactive system [99]. In this vein, the intangible aspect of games as an interactive system should also be recognized, including the flow of the game, the experience, and the emotional/social benefits for participants. In short, in terms of promoting the preservation and sustainable development of culture in peripheral areas, such gamified products should be considered as complementing, not contradicting, cultural tourism.

5.2. Gamification Strategy for Cultural Preservation and Sustainability in Peripheral Areas

With the design of the Tanka ballads game, the gamification of cultural content is demonstrated as a feasible way to raise awareness about and facilitate the preservation of culture, and therefore contribute to local sustainable development.

Further, managerial implications and suggestions for adopting gamification as a strategy for cultural preservation and sustainability in peripheral areas are summarized below:

1. Cultural experience is arguably the primary target of preservation. It is also at the core of cultural innovation and provides the aesthetic for the game to be developed. Therefore, when extracting cultural content at the very beginning of a project, designers need to achieve consensus with local stakeholders on what to preserve.
2. In the same vein, instead of introducing novel forms of interaction, designers should search for interaction that channels user behaviors to resemble the selected cultural content. More specifically, systematic mapping between mechanisms and cultural content is still lacking. Thus, deciding on a game mechanism is reliant largely on the designer's experience. Other tactics include incorporating game designers/experts or turning to collections, such as boardgamegeek.com, for inspiration.
3. In terms of game dynamics, the "gamification of culture" actually has a similar dilemma to games for learning, which is balancing the need to cover the subject matter with the desire to prioritize gameplay [99]. As cultural innovation must first be self-sustaining, it is strongly suggested that the game must at least be smooth and fun.

6. Conclusions

A comprehensive discussion was conducted based on a theoretical analysis of cultural preservation and local sustainable development in association with cultural game development data.

Firstly, whether from the viewpoint of CCIs per se or the trend of glocalization, culture is undoubtedly a core resource for innovation. Therefore, when considering the sustainable development of peripheral areas, local culture with unique characteristics should be considered as the source of distinctive competitiveness, and the intertwined economic system and local culture should be considered under the precondition of prioritizing cultural inheritance in peripheral areas [33–35] to determine a suitable cultural innovation strategy.

Secondly, there is no denying that cultural tourism would still play a vital role, and a properly planned long-term tourism activity is critical for the regional planning to reach the desired living conditions capable of attracting new residents in the future [100]. Nevertheless, the purchase of regional goods (and services) is given the highest priority [100]. This would include tangible cultural products that could strengthen cultural patrimony for the local community and preserve cultural heritage for the greater society.

In this vein, the presented study investigated the Tanka culture in southern China with a gamification approach, took part in its cultural content, and transformed it into a game. At the core of this proposed strategy is the cultural content, which, when transformed into a game, becomes a form of media that can contribute to the preservation and dissemination of culture. From the perspective of game theory, there are unique advantages to such a transformation, both on a behavioral and a conceptual level. On a behavioral level, as mentioned above, the players must have appropriate epistemic access to the goals and rules and commitment prior to the start of the game so that they can be guided by the game's goals and rules [67–70]. Therefore, when cultural behavior, such as a certain ceremony, is transformed into a game, and when people decide to play the game, the players understand and accept the goal of the cultural behavior and are willing to behave in a way that conforms to the inherently cultural rule. Therefore, the game becomes a useful tool, which allows consumers to break free from the boundaries of their own culture, making it easier for them to accept the unique existence of other cultures and possibly further appreciate the development or decline of such cultures. Thus, it can be perceived as a cross-cultural product contributing to cultural preservation in peripheral areas.

On a broader conceptual level, games are cultural products; therefore, they belong to libraries as much as music or movies. They are inherently artifacts of culture through which cultural roles, values, and knowledge bases are transmitted. They both reflect and simultaneously work to reproduce culture [101,102]. It is within the grand processes of innovating and industrializing culture that the transformation of audiences into practitioners, or active cultural participation, can be observed, where the active character of cultural participation goes beyond the passive absorption of cultural stimuli, motivating individuals to make use of their skills to contribute to the process [19].

With unique advantages, design transformation for culture can be achieved via gamification, where existing cultural elements, such as implements, behavior, rituals, and the like, could be properly packaged by conforming to contemporary values or aesthetics. Furthermore, it is through gamification that individuals can be transformed into different types of players, e.g., observers, apprentices, spectators, and performers, who vary in terms of the extent to which they actively or passively participate in the gamified experience and in the degree to which they are absorbed or immersed in the gamified environment [103]. In summary, games provide hedonic value, while at the same time they invite consumers to be involved in broader value co-creation, thereby contributing to the cultural preservation of peripheral areas.

This research contributes to both theory and practice by developing the theoretical background and conducting an early case study. This work advances the endeavor for sustainability through cross-cultural design and showcases how design thinking that goes to the level of industrial logics can lead to potential design strategies. Interestingly, this

research also points to a direction for future research about the fundamental role and potential of materiality for the sustainability of culture and society.

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